Practical 4

20% contribution to the final mark

4 questions, each has 5 points; in total $4 \times 5 = 20$ points

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Assignments

Q1 : What is the relation between the Tweedie distribution and the Poisson-gamma distribution? Write the final model based on the estimates provided in Lab8.

Q2 : In a study of sheets of building materials the permeability of three sheets was measured on three different machines over nine days, for a total of 81 sheets, all of equal thickness. Each measurement is an average permeability of eight random pieces cut from each of the 81 sheets.

Use library(GLMsData); data(perm) to get the data.

1. Determine an estimate of ξ using tweedie.profile(). What EDM is suggested?
2. Fit a suitable Tweedie GLM ensuring an appropriate diagnostic analysis.

Q3 : Consider the respiratory illness dataset in Lab10. Run glm using

$$glm(outcome \sim center + treat + age + baseline,\\ data = respiratory, family = binomial())$$

Analyze the output and interpret the estimates using odds ratio, and compare the results with the output of gee.ind in the Lab10.

Q4 : In Lab11, from the output, the estimate of the scale parameter is equal to 1.015512. Thus, it is close to one and we may think of redoing the analysis by setting it to 1. Use the amendment "scale.fix = TRUE, scale.value = 1", and compare the outputs. What is the change if you apriori set the scale parameter to 1?